THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which pair of materials would make the best flask lid? 1

(a) iron and copper **(b)** glass and steel **(c)** silver and gold **(d)** cork and plastic

2. A metal bowl on a table loses heat to the cold floor by: 1

(a) conduction **(b)** convection only **(c)** evaporation **(d)** radiation only

3. Which activity illustrates only radiation? 1

(a) feeling warmth from a lit bulb in a closed glass jar
(b) stirring soup
(c) boiling water
(d) heating a pan

4. A student wants a quick and even heating of water. She should: 1

(a) not heat **(b)** heat it from above
(c) heat it from the sides only **(d)** heat it from below

5. Why does the water in the middle layers of a heated beaker move in circular paths? 1

(a) conduction **(b)** evaporation
(c) radiation **(d)** convection currents

6. A pan on a stove loses heat from its sides by: 1

(a) evaporation only **(b)** convection and radiation
(c) melting **(d)** conduction only

7. A student suspects convection is absent in a vacuum. Which statement is true? 1

(a) convection is conduction **(b)** convection is radiation
(c) convection needs no medium **(d)** convection needs a fluid

8. In a vacuum flask, which transfer is still partly possible without special silvering? 1

(a) convection **(b)** evaporation **(c)** radiation **(d)** conduction

9. A blacksmith holds a red-hot iron with tongs because: 1

(a) iron is shiny **(b)** wood is cold
(c) iron conducts heat to his hand **(d)** iron is hard

10. Which of the following is the best reason for placing a heater near the floor? 1

(a) it is hot **(b)** it saves space
(c) floor is cold **(d)** warm air rises and spreads

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does smoke from a cigarette rise in still air? 2

12. Why does the tin roof of a hut become very hot in sunlight? 2

13. Why is the roof painted white in many hot-climate houses? 2

14. Why is a metal rod with a wooden handle used for a hot-wire cutter? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why do people in cold regions build igloos with snow blocks and a small entrance? 3

16. Why does water in a metal bucket remain cooler in a shed than outside in the sun? 3

17. Explain why dust particles seen in a sunbeam in a room move. 3

OR

Explain why a pressure cooker cooks faster and how the lid and the weight help. 3

18. Describe the movement of air in a room with a window open at the top and bottom on a cold day. 3

OR

Why do some animals like elephants flap their ears? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Analyse why a tea kettle's handle is made of plastic and the body of metal, whereas a cooler box is made of plastic and foam. 5

OR

Explain the transfer of heat in a heated metal rod, a pot of water and the Sun-Earth system, using the terms conduction, convection and radiation. 5

20. Describe how to demonstrate that radiation can pass through glass but conduction needs contact. 5

OR

Prepare a note on energy saving at home using ideas of heat transfer. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND